

# ANMOL SAHOO

anmol.sahoo25@gmail.com | [anmolsahoo25.github.io](https://anmolsahoo25.github.io) | [github.com/anmolsahoo25](https://github.com/anmolsahoo25)

I am a fourth-year Ph.D. student interested in building programming language abstractions that make it easier to develop correct, expressive and performant distributed systems. I use my skills in language design, compilers, type theory and formal reasoning to build tools in OCaml.

## EDUCATION

**Purdue University** Aug 2020 - Ongoing  
Doctor of Philosophy in Computer Science

**BITS Pilani, Goa Campus** Aug 2013 - Jul 2017  
Bachelor of Engineering in Mechanical Engineering

## RESEARCH EXPERIENCE

**Semantics for Algebraic Effects and Handlers** Under submission

- Proposed a novel framework for extending algebraic effects and handlers
- Developed the formal semantics of proposed framework
- Demonstrated applications of the technique to solve real-world problems

**Verification of Weak Memory Libraries** Under submission

- Developed an encoding in Z3 for automated verification of weak-memory libraries
- Verified open-source concurrent, lockfree libraries such as Folly and ring-buffer queues

**Retrofitting Parallelism onto OCaml** ICFP 2020  
KC Sivaramakrishnan, Stephen Dolan, Leo White, Sadiq Jaffer, Tom Kelly, Anmol Sahoo  
Sudha Parimala, Atul Dhiman, Anil Madhavapeddy

- Implemented a poll insertion pass to reduce pause times and prevent deadlocks in the GC
- Assisted with compiler backend fixes required for upstreaming multicore OCaml

## WORK EXPERIENCE

• Purdue University, Graduate Research Assistant Aug 2020 - Ongoing  
• IIT Madras, Research Engineer Jun 2018 - Jul 2020  
• Johnson Controls, Engineering Trainee Aug 2017 - May 2018

## SKILLS AND COURSES

- Languages: OCaml, C, Python, Coq
- Tools: Z3, perf, gdb/rr, Linux
- Courses: Programming Languages, Reasoning about Programs, Algorithms and Complexity
- Open Source: Dynamic race detection for OCaml, ScyllaDB driver for OCaml

## EXTRACURRICULAR ACTIVITIES

- Teaching assistant for Programming Languages and Programs and Proofs at IIT Madras
- Placement coordinator for 600 graduating students at BITS Pilani
- Long-distance running - 10 completed half marathons with 2:04 best time